Grade 6 Pre-Assessment

Do your best. It's okay to skip a problem you don't know yet. Score: _____ / 35

Part 1 Ratios

1) $6 : 1 = \square : 4$

2) $9 : 2 = \square : 6$

Part 2 Rates

3) 150 miles in 5 hours = ☐ miles per hour

4) 72 pages in 6 days = ☐ pages per day

Part 3 Percents

5) 20% of 85 = ☐

6) 25% of 44 = ☐

Part 4 Dividing fractions

7) $\frac{5}{8} \div \frac{1}{4} = \frac{\square}{\square}$

8) $\frac{2}{3} \div \frac{1}{9} = \square$

Part 5 Long division

9) $3{,}404 \div 23 = \square$

10) $6{,}370 \div 26 = \square$

Part 6 Decimal operations

11) $16.2 \div 0.9 = \square$

12) $9.75 \div 1.5 = \square$

Grade 6 Pre-Assessment (continued)

Part 7 GCF and LCM

13) GCF of 50 and 80 = ☐

14) LCM of 4 and 7 = ☐

Part 8 Negative numbers Write < or >.

15) 5 ☐ −5

16) −2 ☐ −12

Part 9 Expressions

17) when n = 3: 7n + 8 = ☐

18) when n = 10: 4n + 3 = ☐

Part 10 One-step equations

19) x + 26 = 43: x = ☐

20) 9x = 99: x = ☐

Part 11 Area of shapes

21) triangle, base 13 in, height 8 in: area = ☐ sq in

22) triangle, base 9 m, height 6 m: area = ☐ sq m

Part 12 Volume with fractions

23) $1\frac{1}{2} \times 3 \times 3 = \frac{\square}{2}$

24) $2\frac{1}{2} \times 2 \times 6 =$ ☐

Part 13 Surface area

25) box 5 × 3 × 4 cm: surface area = ☐ sq cm

Grade 6 Pre-Assessment (continued)

Part 13 Surface area (continued)

26) **box 9 × 4 × 2 ft: surface area = ☐ sq ft**

Part 14 Mean and median

27) **median of 7, 15, 9, 13 = ☐**

28) **mean of 5, 8, 6, 9 = ☐**

Part 15 Data

29) **12, 13, 13, 14, 48: mean = ☐ and median = ☐**

Part 16 Word problems

30) A car goes 336 miles in 6 hours. How many miles does it go in 1 hour?

Answer ☐ miles

31) Juice is mixed 2 cups of mango for every 5 cups of water. How much water goes with 6 cups of mango?

Answer ☐ cups

32) 6 bottles of water cost $4.20. How much is 1 bottle?

Answer ☐

Grade 6 Pre-Assessment (continued)

Part 16 Word problems (continued)

33) A school buys 2,016 pencils in boxes of 24. How many boxes?

Answer ☐ boxes

34) A $45 shirt is 20% off. How many dollars do you save?

Answer ☐

35) You have 3/4 of a pizza. Each slice is 1/8 of the pizza. How many slices is that?

Answer ☐ slices